PROJET IAR 2025-2026

Models of Hippocampally Dependent Navigation, Using The Temporal Difference Learning Rule (2000)
Morris, Foster, Dayan

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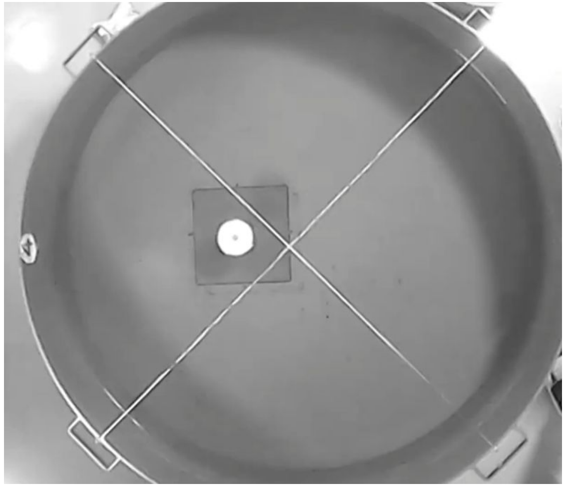
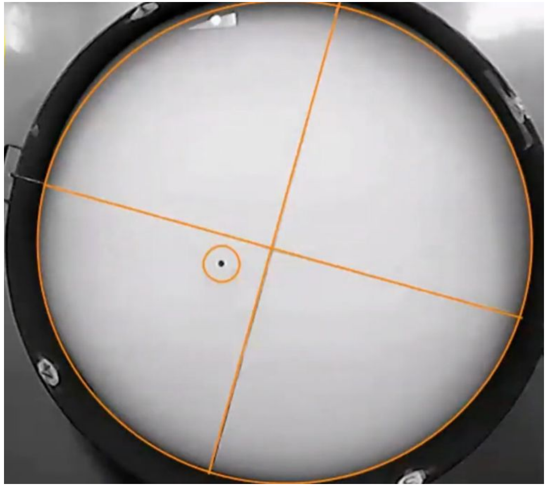
Morris Water Maze

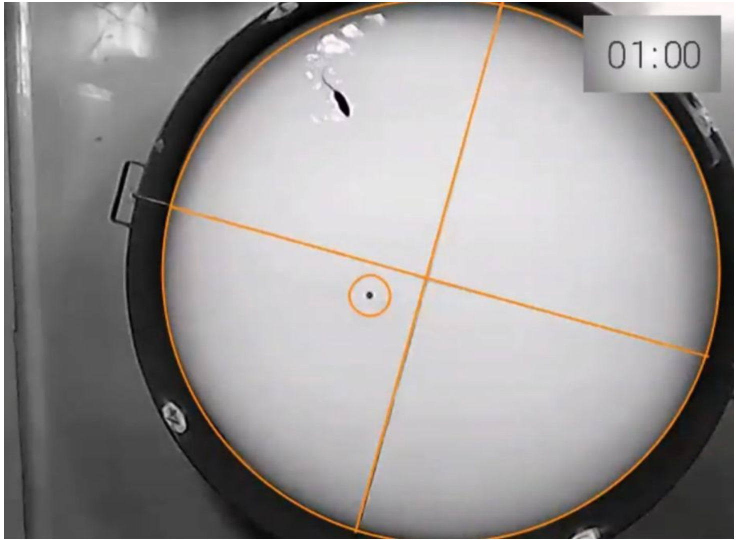
La piscine est rendue opaque pour que la plateforme soit invisible.

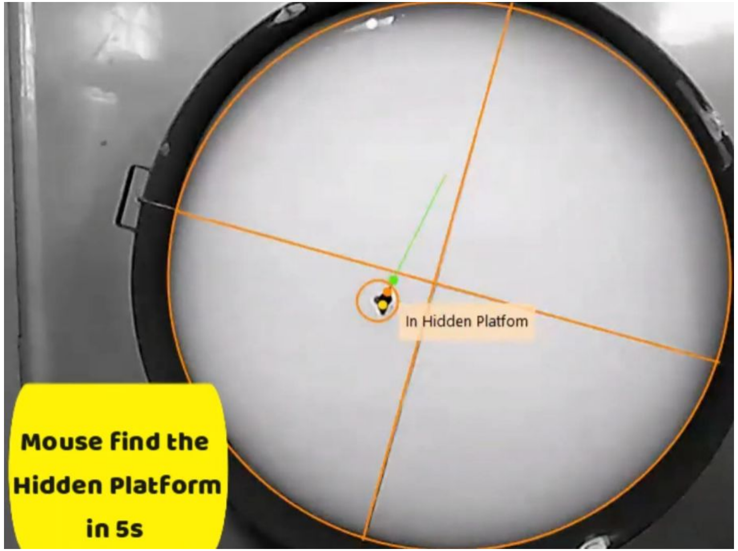




Morris Water Maze

Le rat est lâché dans la piscine depuis un bord, et explore l'environnement jusqu'à trouver la plateforme

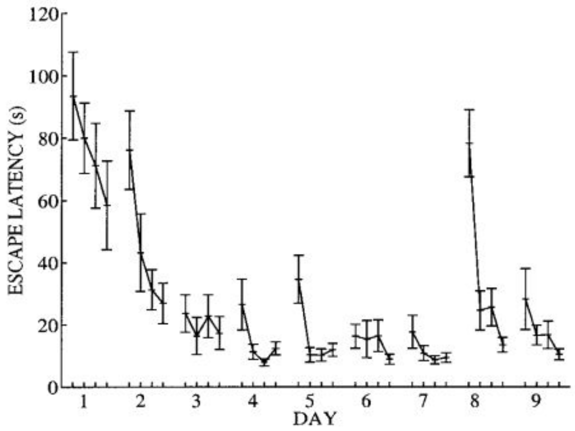




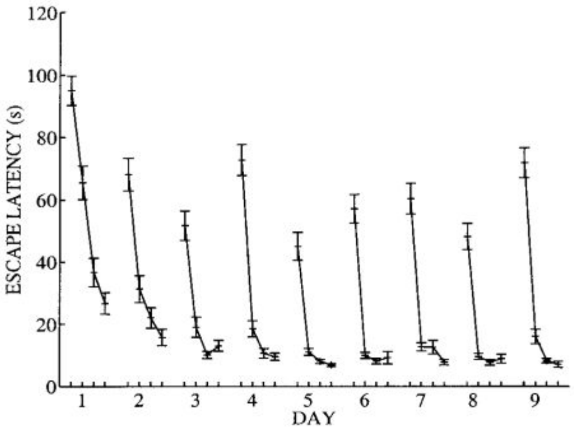
Morris Water Maze

En moyenne, le rat est beaucoup plus performant les essais suivants, ayant appris la position de la plateforme. Figure 1 (a et b)

a RMW



b DMP



Steele and Morris (1999)

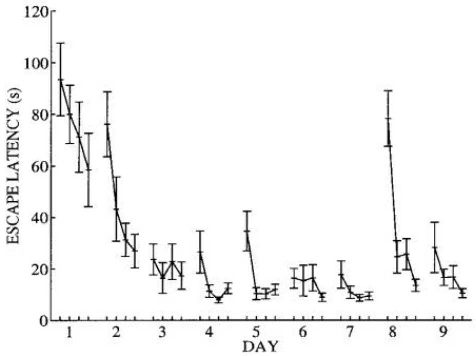
Morris Water Maze

Il est essentiel de noter deux points d’attention pertinents :

1) **Les pics lors du déplacement de la plateforme**

2) **La tendance globale**

a



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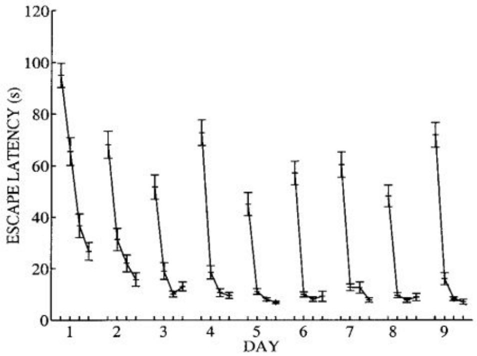
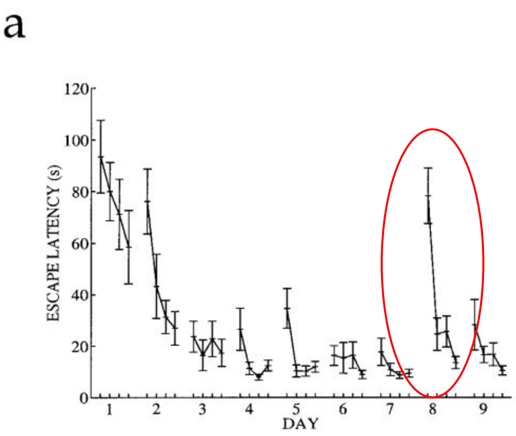
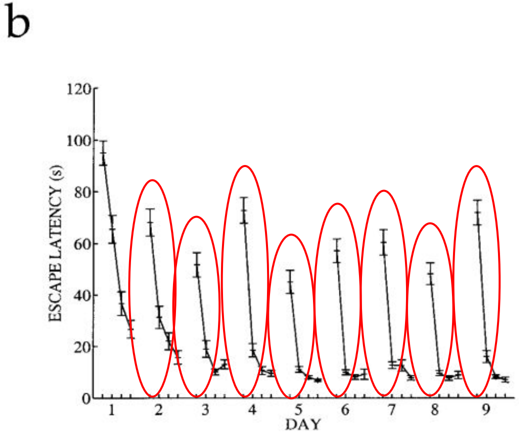


Morris Water Maze

1) Les pics lors du déplacement de la plateforme

La plateforme est déplacé chaque jour dans la tâche DMP, et seulement le 8e jour de la tâche RMW. Dans ces étapes, le rat doit bien sûr explorer à nouveau pour la retrouver, ce qui cause les pics.



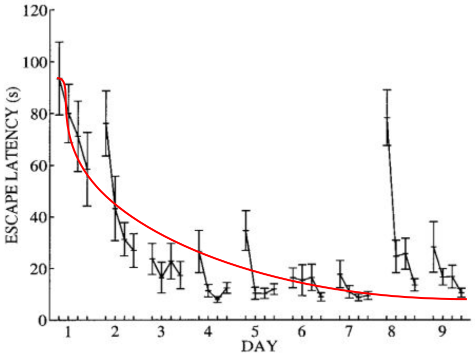


Morris Water Maze

2) La tendance globale

Même avec les pics, dans les deux tâches on voit cette tendance à la baisse au fil du temps. Ce qui suggère que le rat possède une représentation interne de son environnement qui s'améliore avec le temps. (Lignes indicatifs)

a



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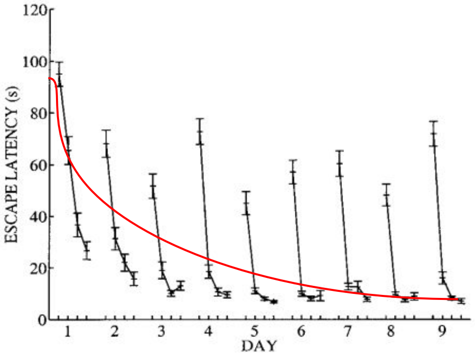
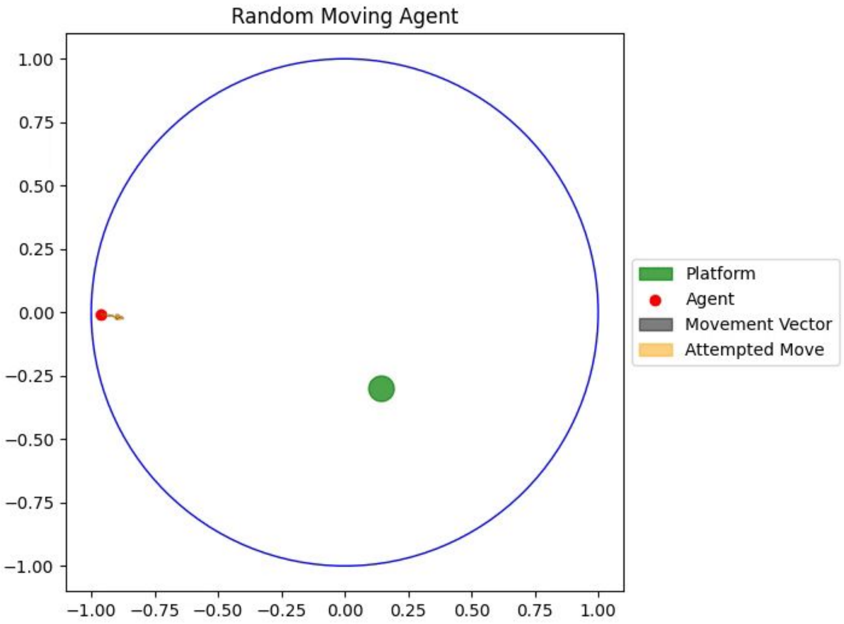


La modélisation / simulation de l’environnement

On simule l’environnement de la piscine et l’agent (rat) en Python



La modélisation / simulation de l'environnement

Principales caractéristiques de la simulation mises en oeuvre :

- Water Maze

Simulation Procedures

We simulated the swimming behavior of a rat in a 2-m-diameter circular watermaze, which contained a 0.1-m-diameter escape platform. These parameters are the same as those in Steele

- Discrétisation du temps

rat bounced off. Any move into the platform area was counted as a move onto the platform. Space was treated as a continuous variable; however, time was discretized into steps of 0.1 s. Simulations with 0.01-s bins produced similar results to those

- Règles

Following the experimental protocols, each trial began at one of four starting locations located at the north, south, east, and west edges of the pool, and ended when either the rat reached the platform, or a time-out of 120 s was reached. For RMW, the

- Tâches

platform, or a time-out of 120 s was reached. For RMW, the platform remained in the same location throughout the simulation. In DMP, the platform was moved to a novel location after every four trials.

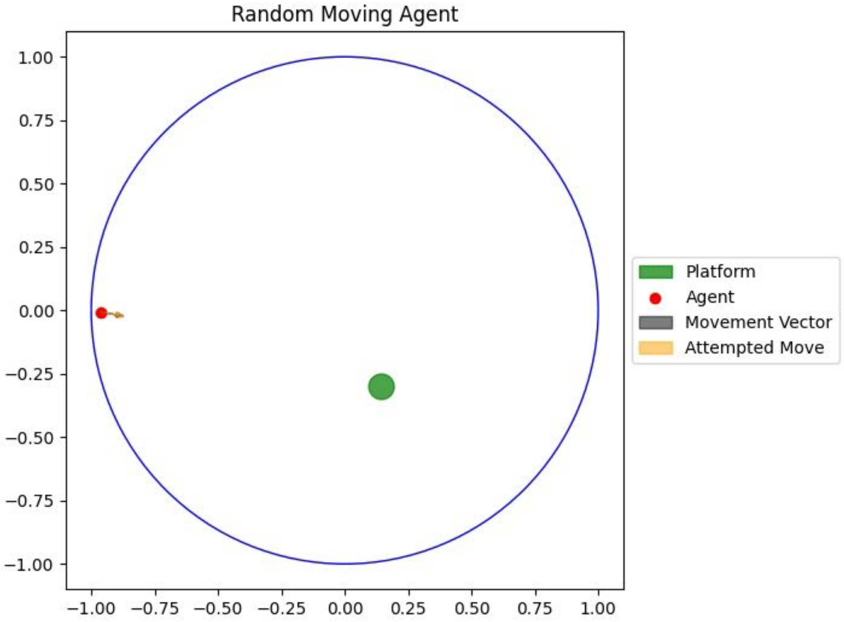
La modélisation / simulation de l’environnement

- Vitesse, Collisions

and Morris (1999). The swimming speed of the rat was constant, at 0.3 ms^{-1}. The walls were treated as reflecting boundaries: the rat “bounced” off. Any move into the platform area was counted

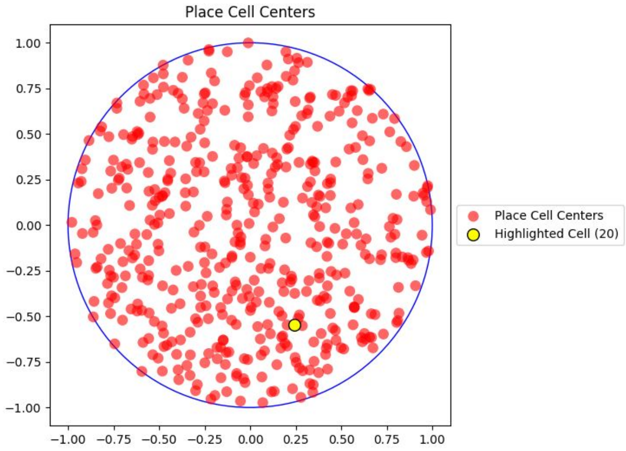
- Momentum

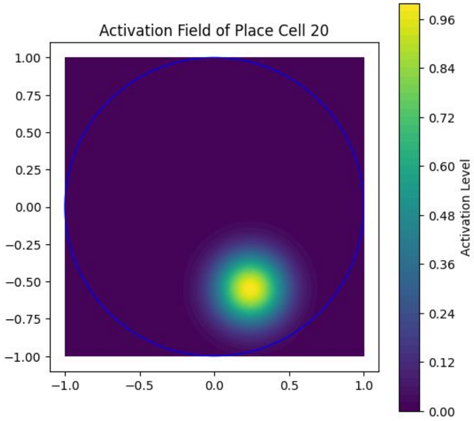
In reality, a rat cannot choose a different direction at the fine-grained time steps of the temporally discrete simulation. To model momentum, the direction the rat heads was given by a mixture of control as specified by the actor, and the previous heading, in the ratio 1:3. This restricts the turning curve of the rat, and is particularly important early on, when the whole pool must

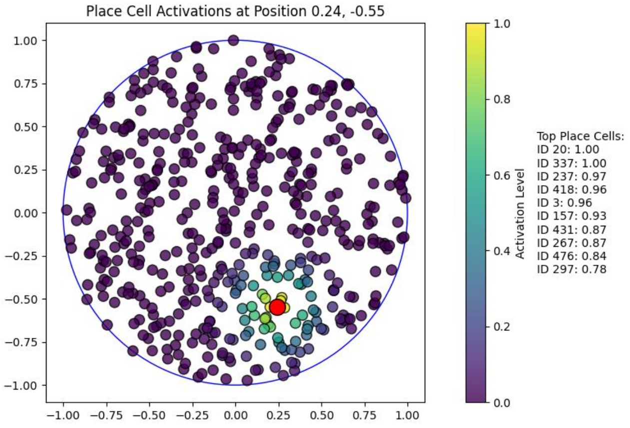


Hippocampal Place Cells

On modélise l’activité des place cells comme des fonctions gaussiens

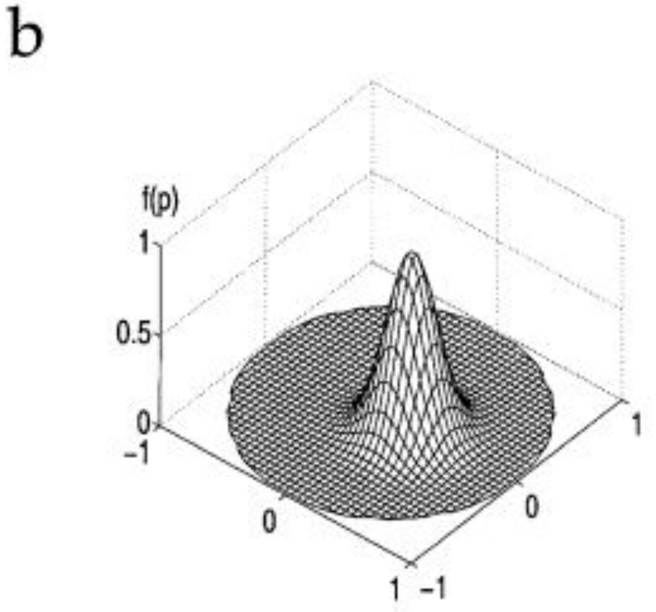


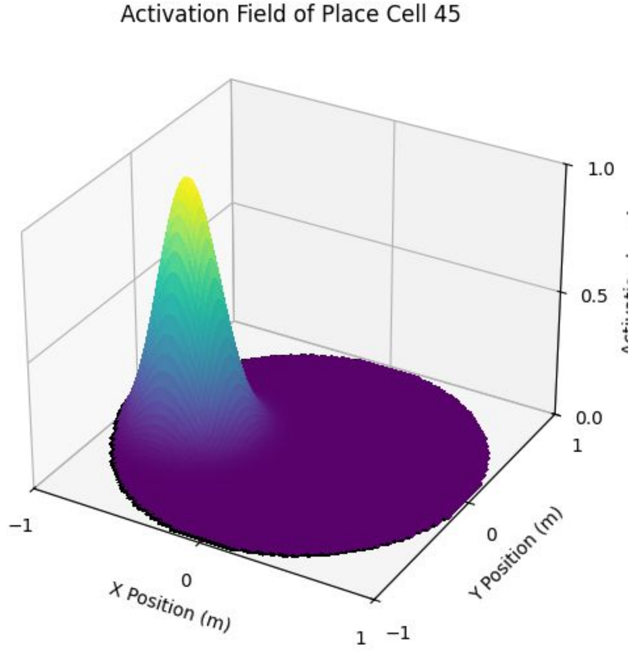




Hippocampal Place Cells

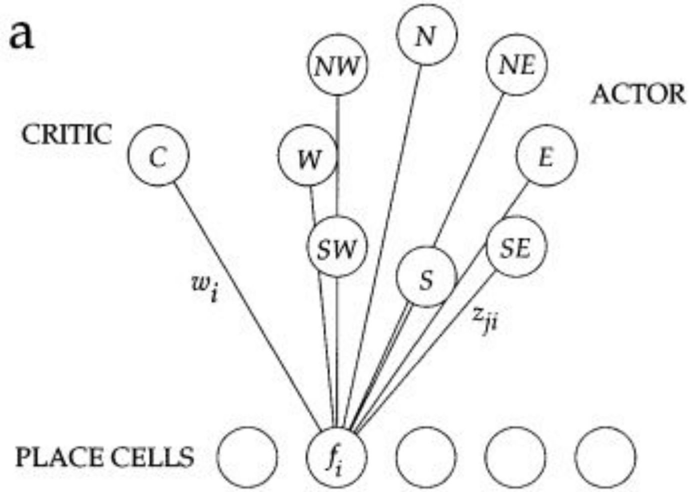
Figure 2b





Système Actor-Critic

Figure 2a



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The Actor

The actor is shown in Figure 2a. For convenience, the rat is allowed to move in one of eight possible directions at each time step (e.g., north, northeast, east), and so the actor makes use of eight action cells a_j, $j = 1 \ldots 8$. Just as in Equation 2, at position p the activity of each action cell is

$$a_j(\mathrm{p}) = \sum_i z_{ji} f_i(\mathrm{p})$$

where z_{ij} is the weight from place cell i to action cell j. This activity is interpreted as the relative preference for swimming in the jth direction at location p: the actual swimming direction is chosen stochastically, with probabilities P_j related to these activities by:

$$P_j = \frac{\exp(2a_j)}{\sum_k \exp(2a_k)}. \tag{9}$$

Page 5

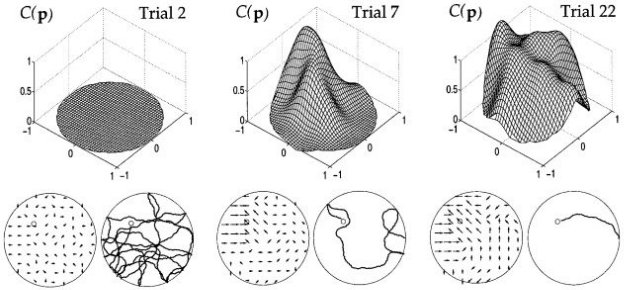
The Critic

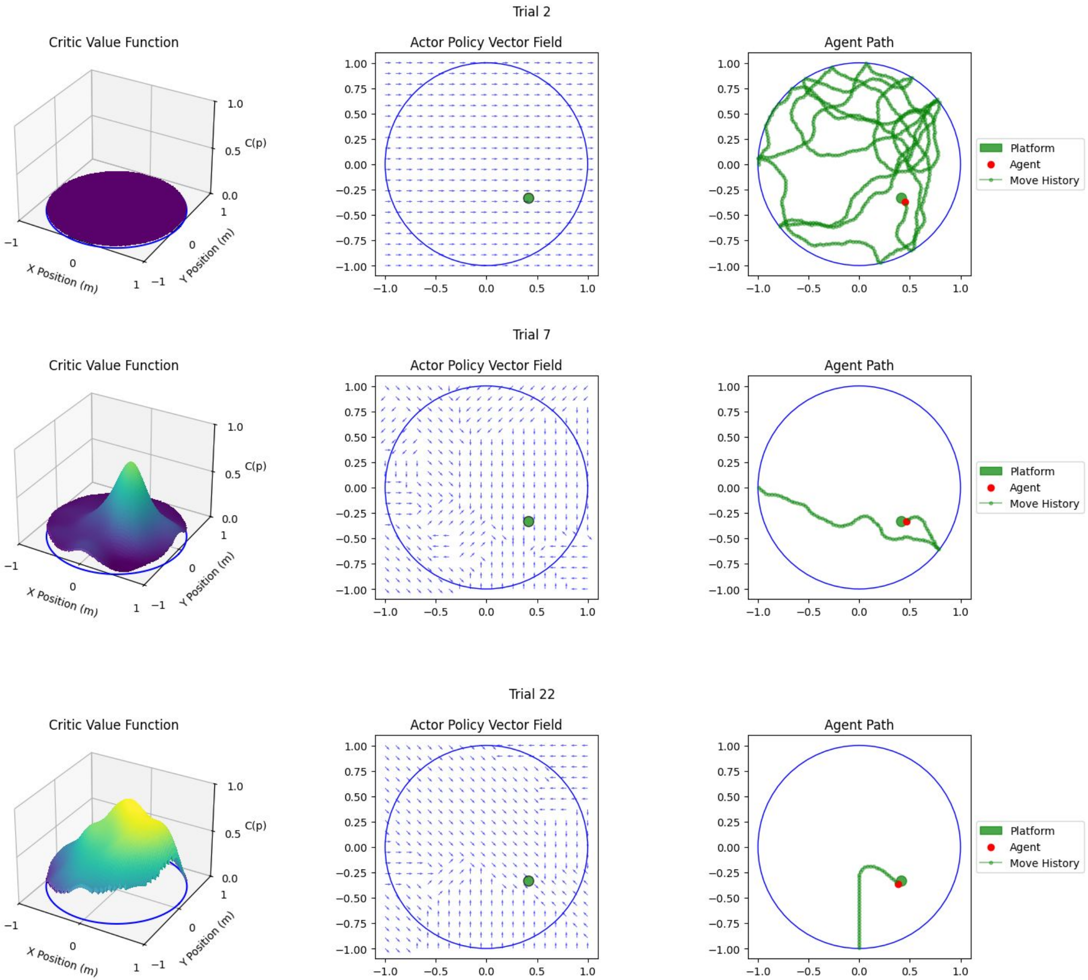
The critic has a single output cell, whose firing rate at a location p is given by a weighted sum of the firing rates of place cell inputs $f_i(\mathrm{p})$:

$$\mathrm{C}(\mathrm{p}) = \sum_i w_i f_i(\mathrm{p}) \tag{2}$$

Système Actor-Critic

Figure 3



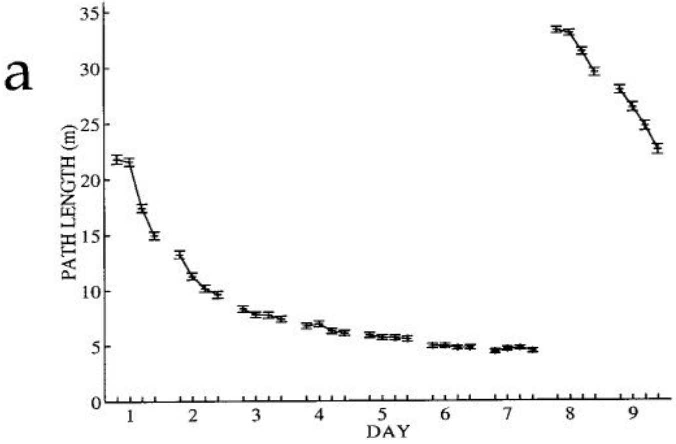


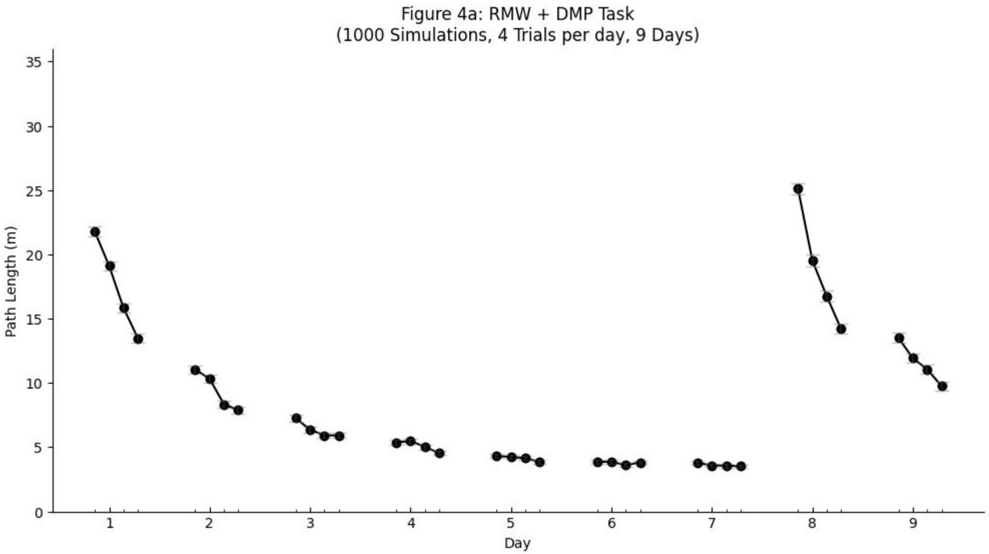
Système Actor-Critic

Figure 4a (RMW)

Pour la tâche RMW, on voit que le modèle apprend comme le rat, produisant des chemins assez directs dans quelques jours (en moyenne).

FIGURE 4. Performance of the actor-critic model. For each data point, the mean and standard error in the mean are obtained from 1,000 simulation runs. (a) RMW task, in which the platform occupies the same location. The actor-critic captures acquisition, producing direct paths after around 10 trials. For the last eight trials, however (days 8 and 9), the platform is moved to a different position (reversal), and the model fails to adapt rapidly enough. These simulation results can be compared to Figure 1a. (b) DMP task, in



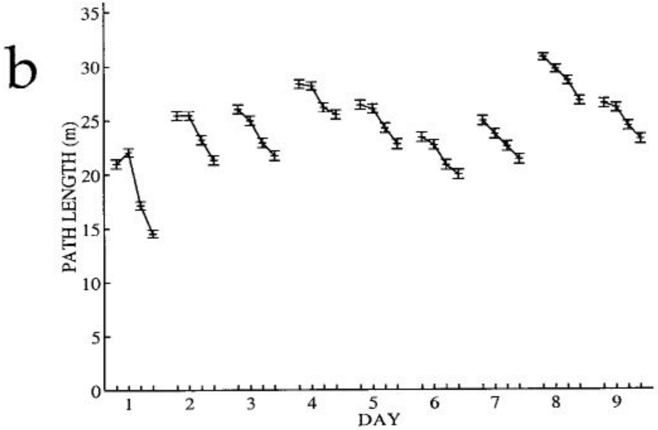


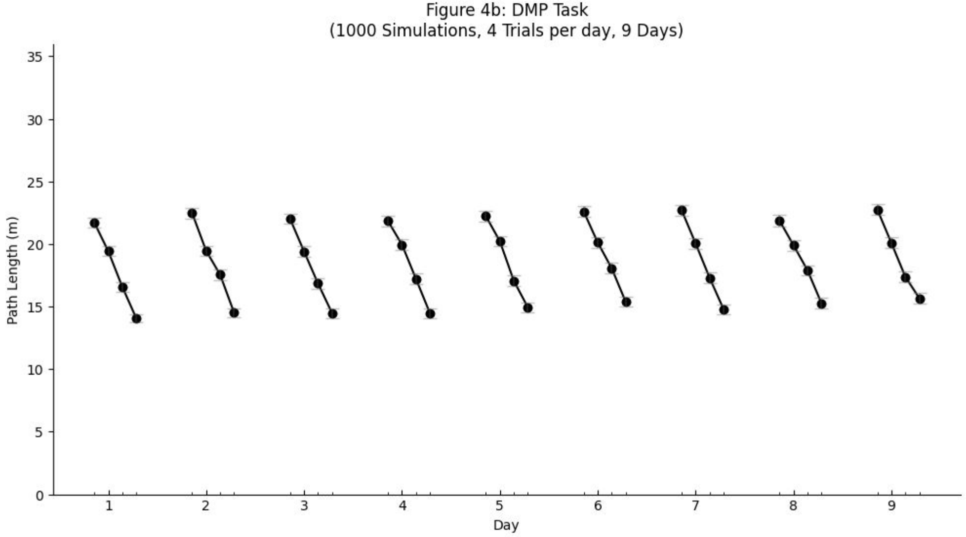
Système Actor-Critic

Figure 4b (DMP)

Par contre, la système Actor-Critic seul ne peut pas correctement modéliser le comportement du rat dans les jours où le plateforme est déplacée.

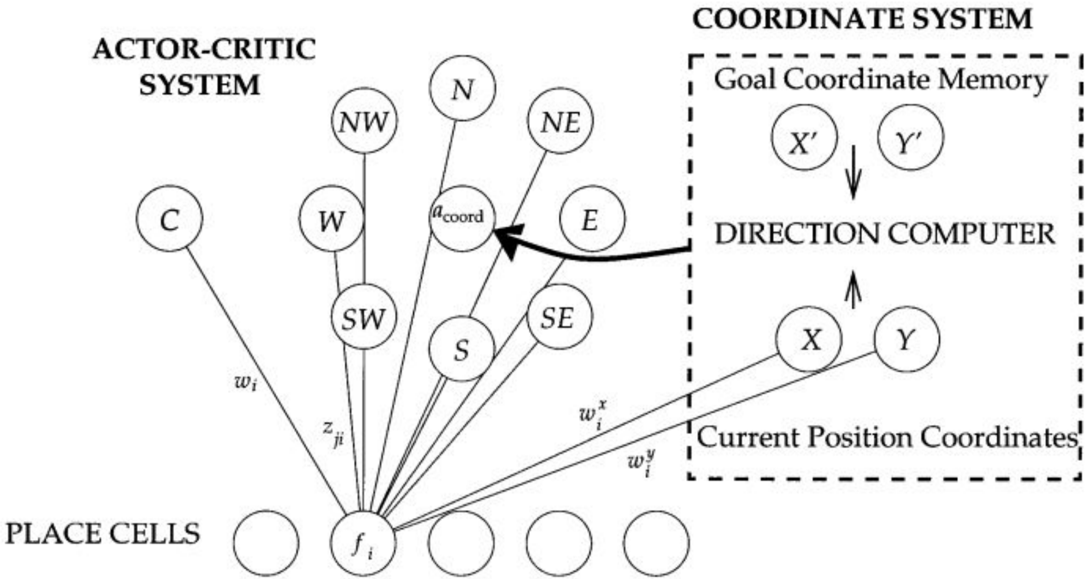
which the platform remains in the same position within a day, but occupies a novel position on each new day. The actor-critic model captures acquisition for the four trials of day 1, for which the task is indistinguishable from RMW. However, as a soon as the platform is moved, the actor-critic not only fails to generalize to the new goal location, but suffers from interference from the previous days' goal locations. Rats suffer neither of these limitations (Fig. 1b).





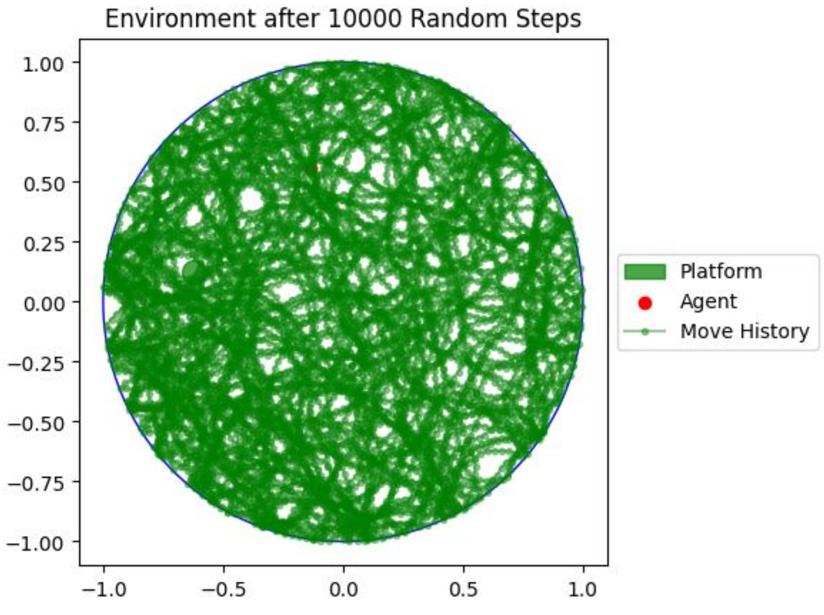
Système de coordonnées

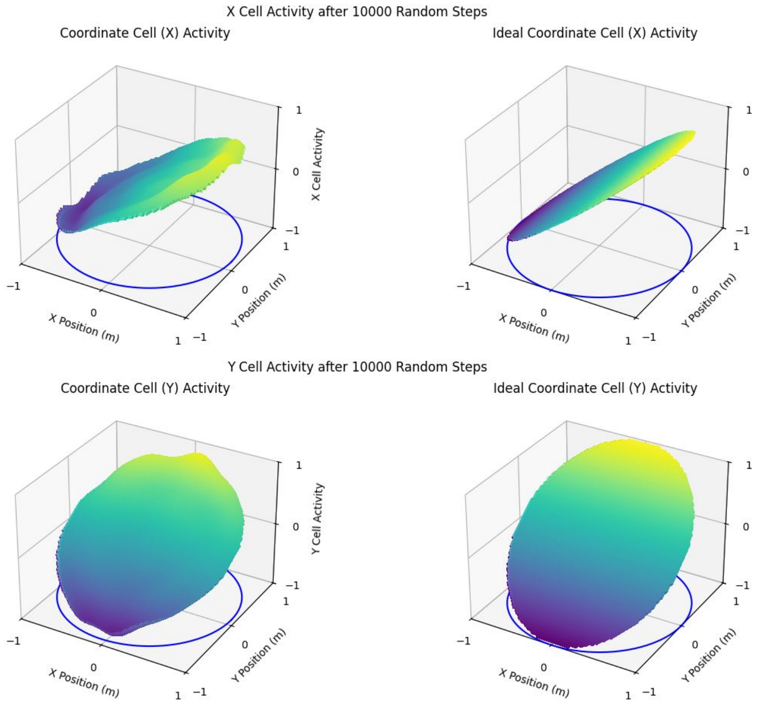
Figure 5



Système de coordonnées

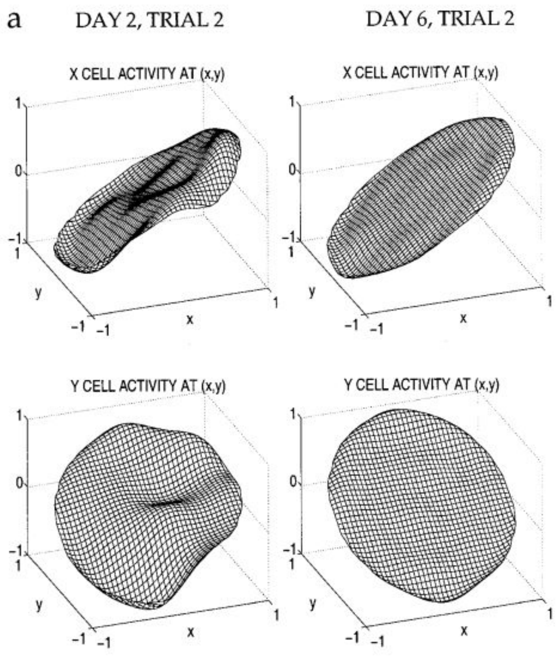
On modélise l'activité des deux cellules coordonnées (X et Y)

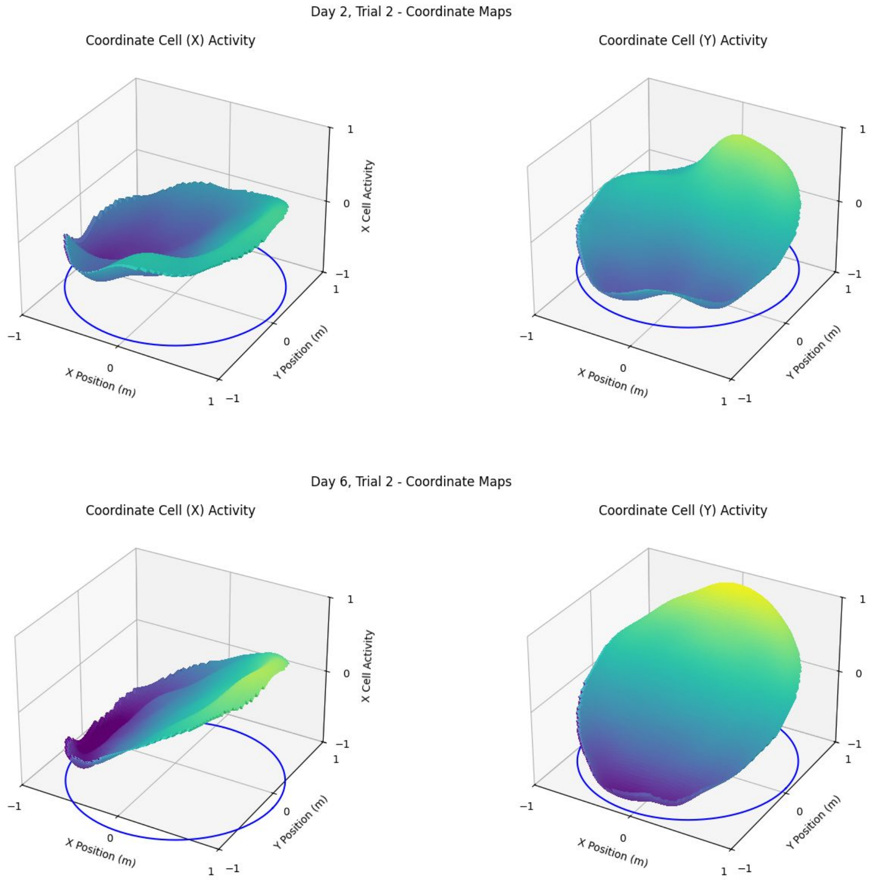




Modèle combiné

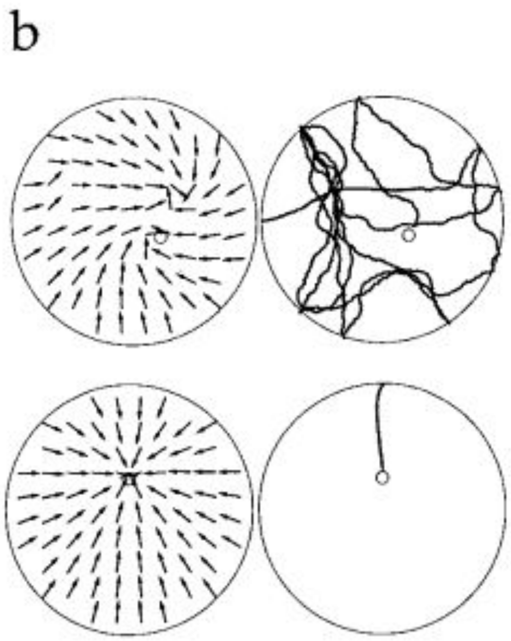
Figure 6a

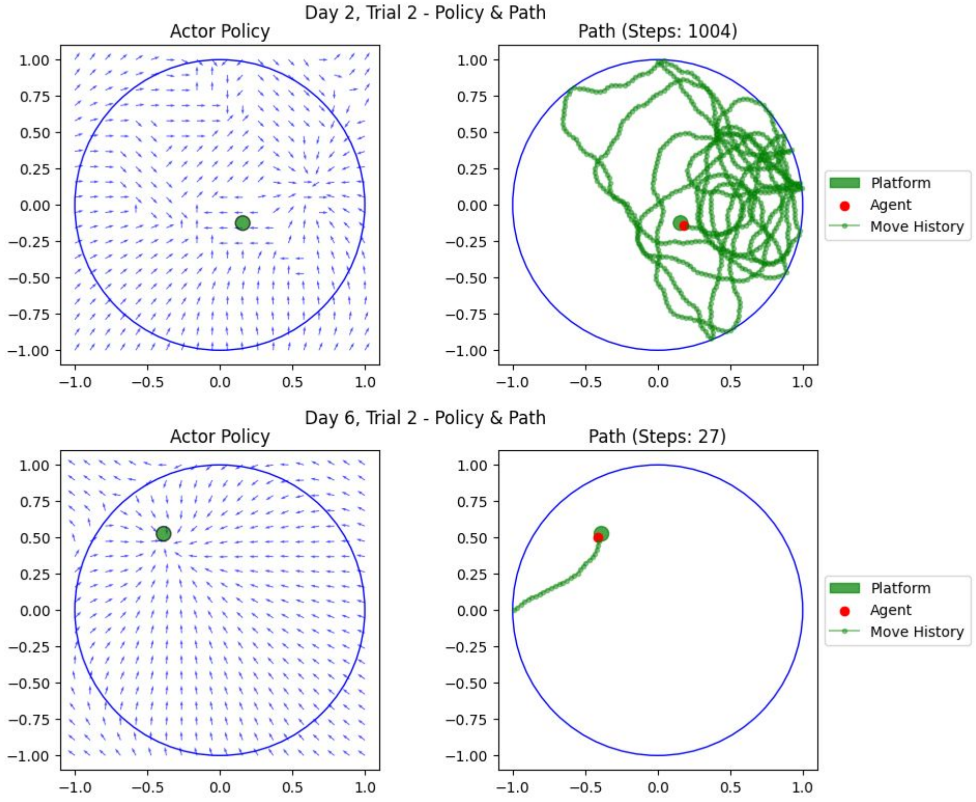




Modèle combiné

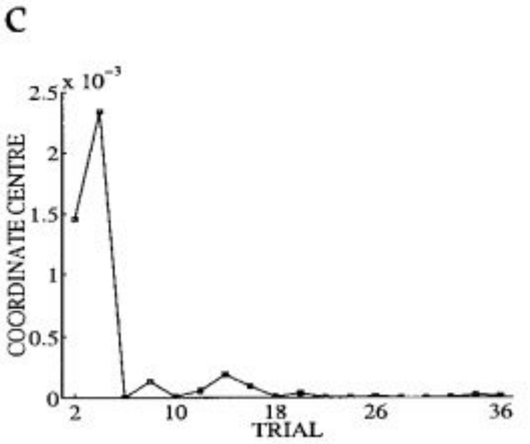
Figure 6b

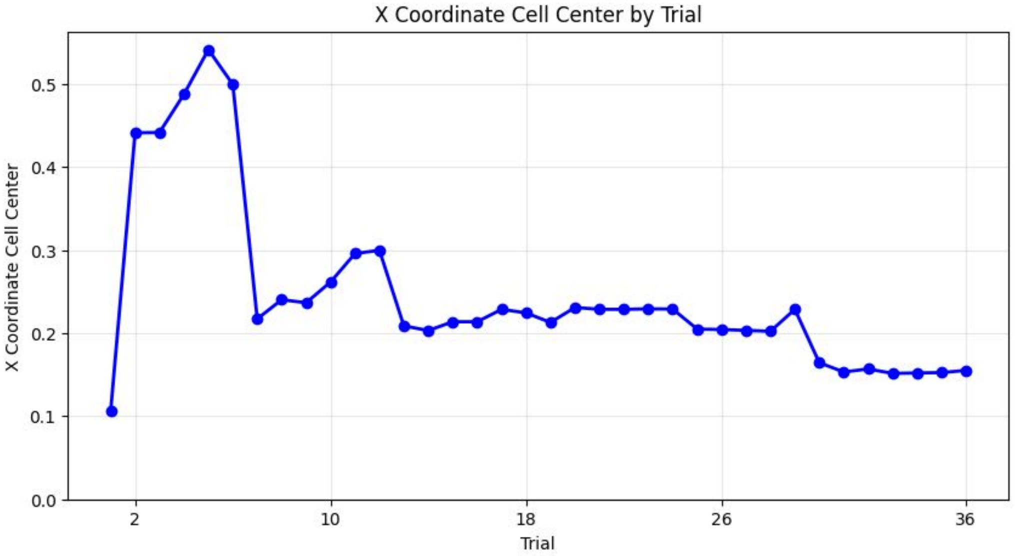




Modèle combiné

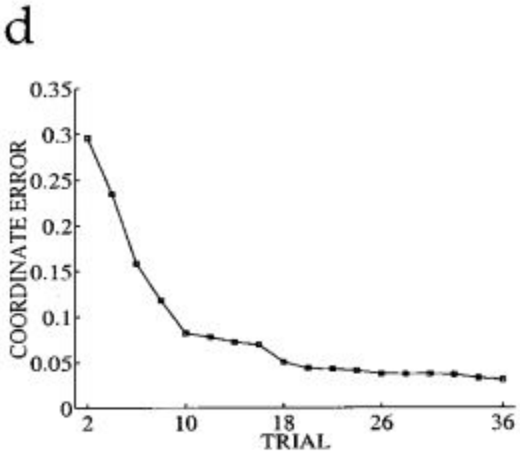
Figure 6c

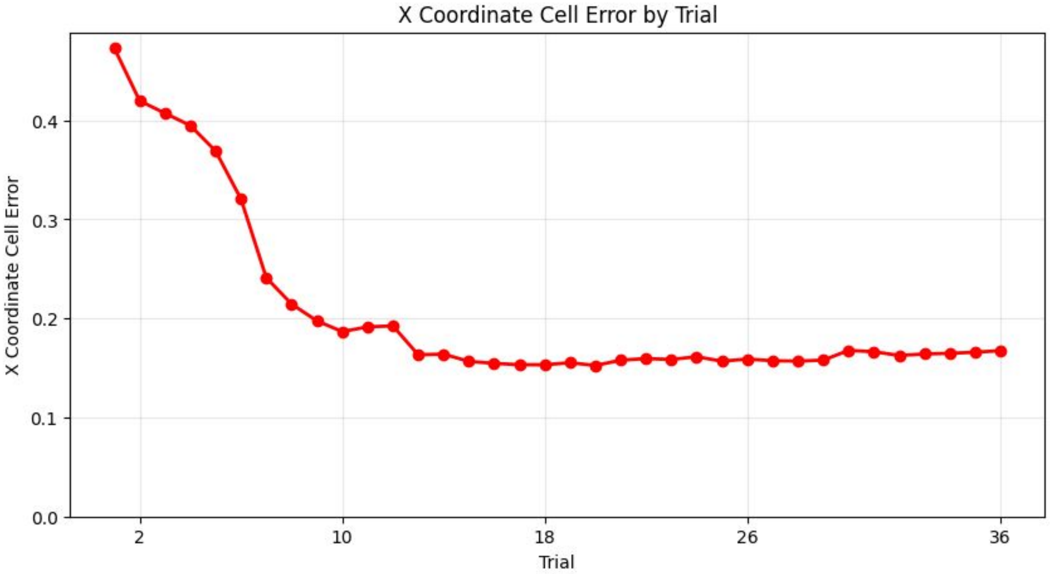




Modèle combiné

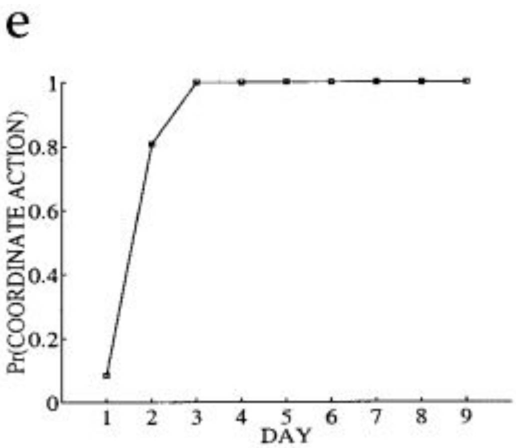
Figure 6d

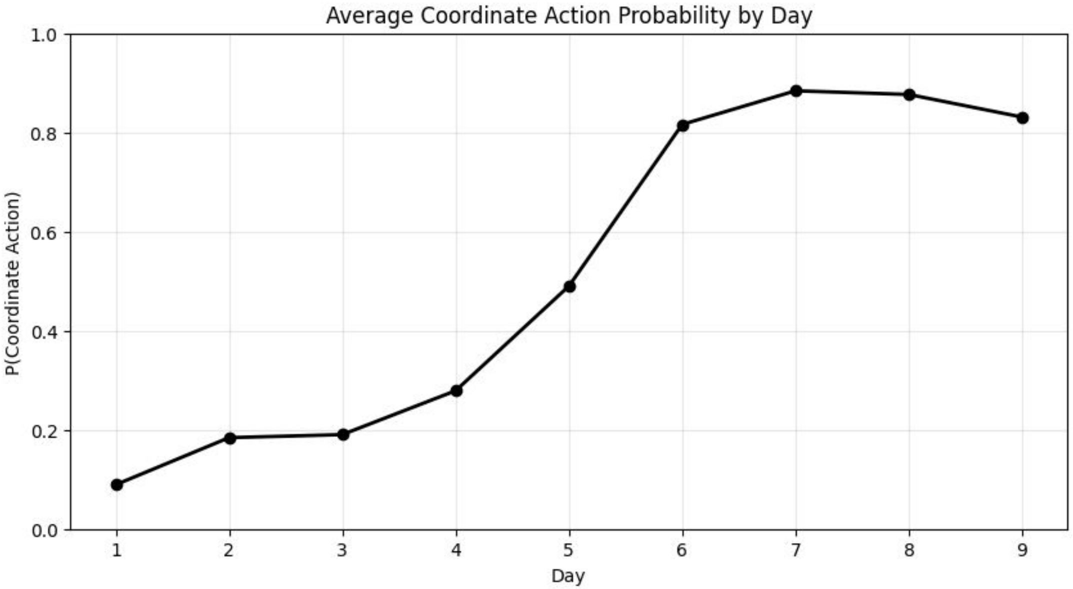




Modèle combiné

Figure 6e





Modèle combiné

Figure 8a (RMW)

On voit que le modèle combiné performe beaucoup plus proche au rat en figure 1 que le modèle Actor-Critic seul en figure 4.

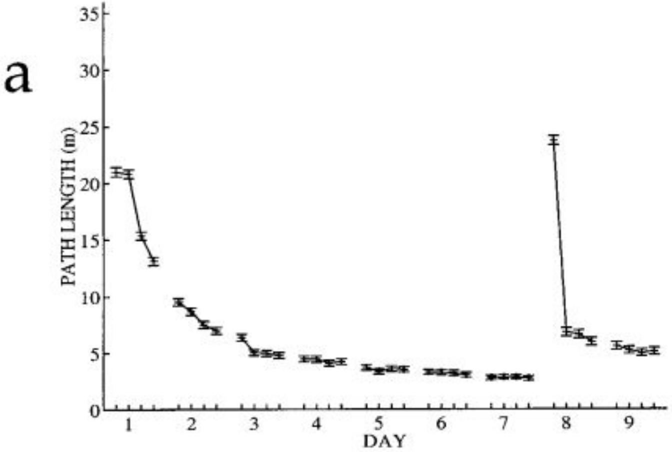
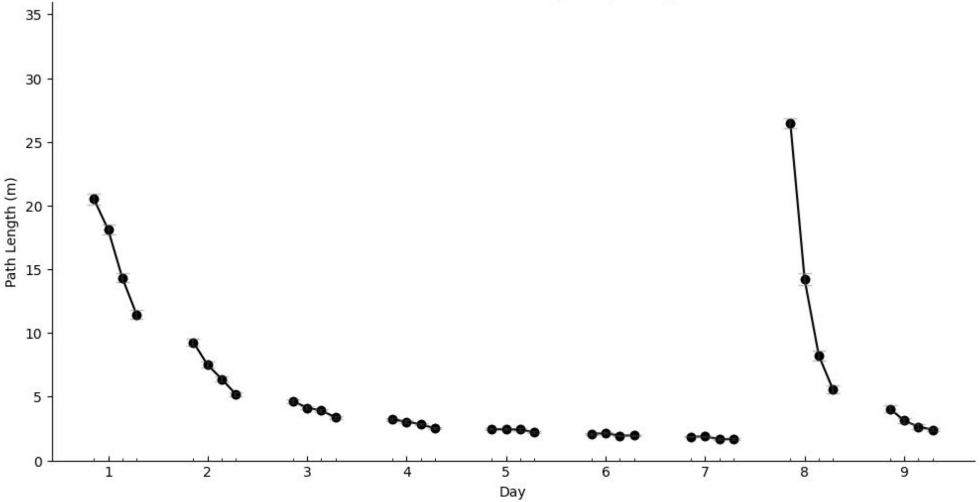


Figure 8a: RMW + DMP Task
(1000 Simulations, 4 Trials per day, 9 Days)



Modèle combiné

Figure 8b (DMP)

On voit que le modèle combiné performe beaucoup plus proche au rat en figure 1 que le modèle Actor-Critic seul en figure 4.

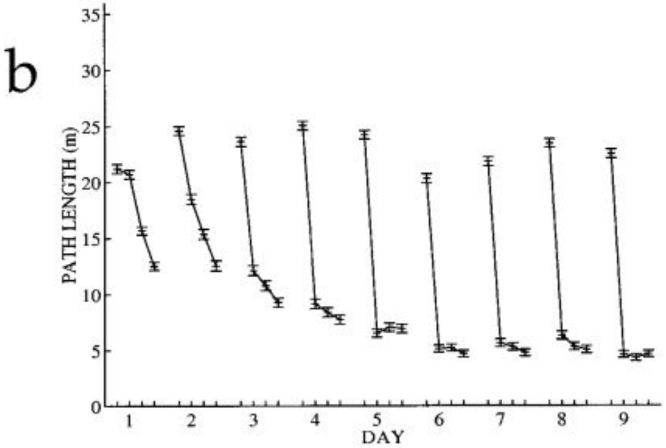
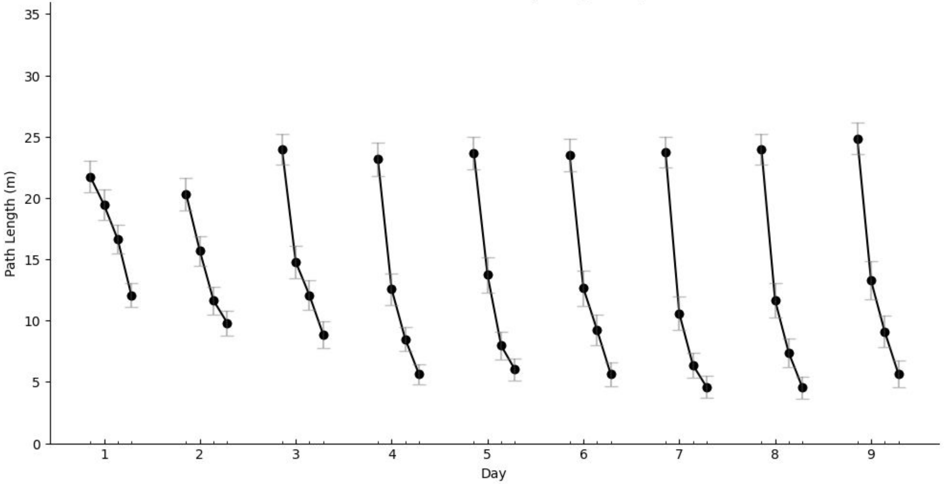
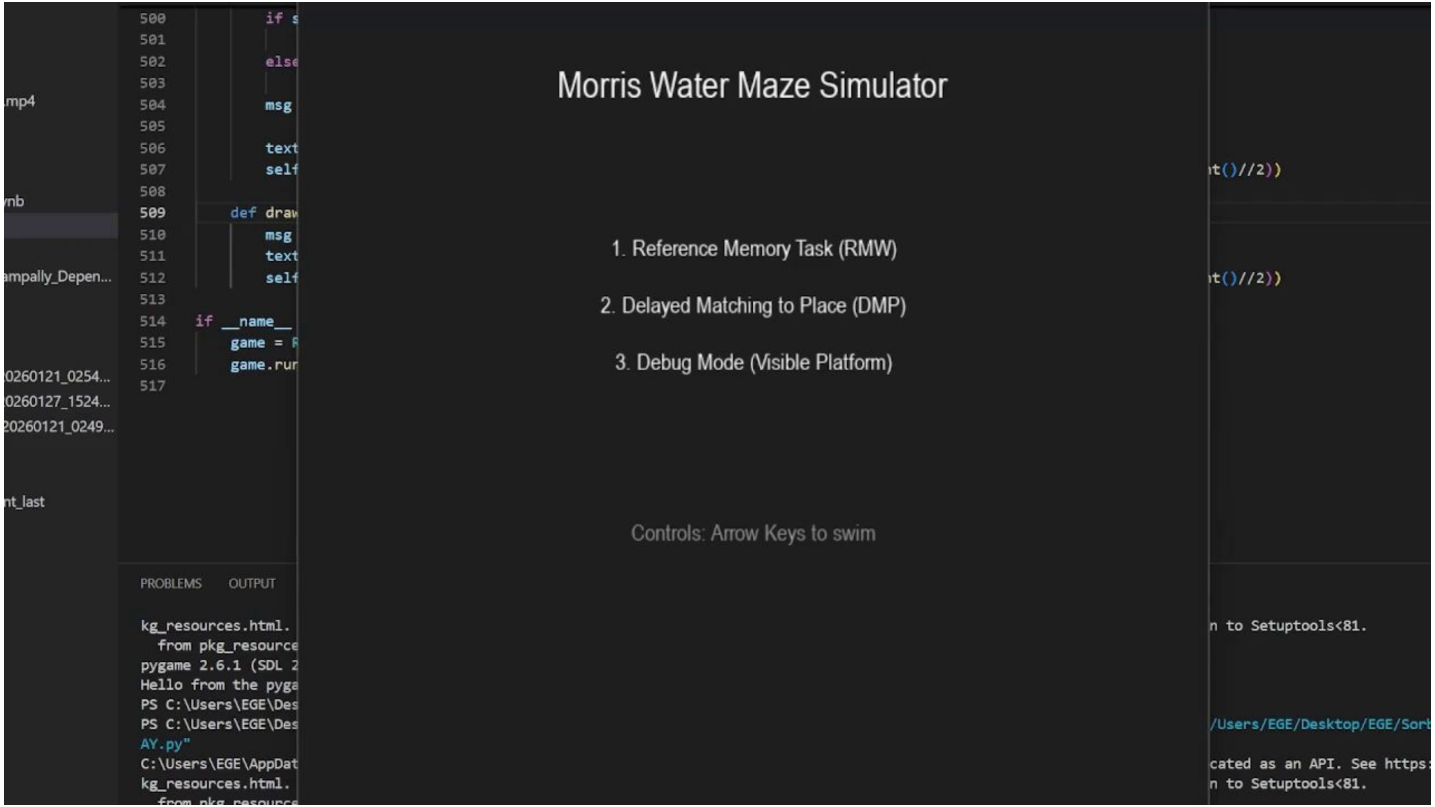


Figure 8b: DMP Task
(100 Simulations, 4 Trials per day, 9 Days)

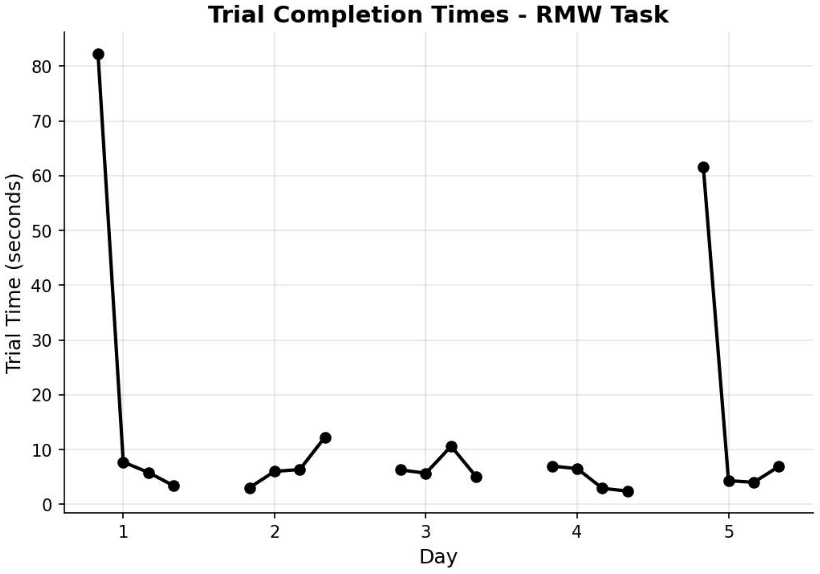


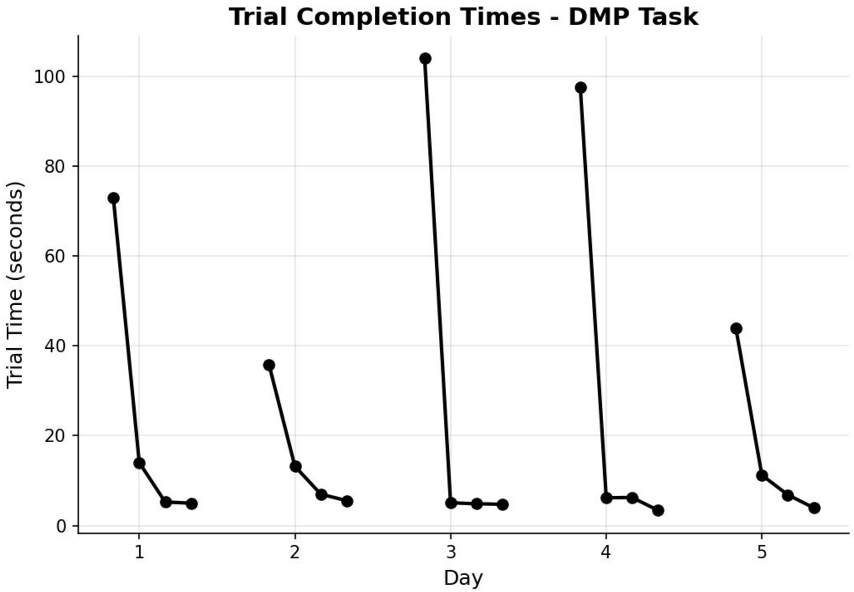
Bonus: Interface Jouable



Bonus: Interface Jouable

Donnés Humains





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